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In [16]: """
This notebook conducts exploratory data analysis (EDA) for the Histopathology dataset.
The dataset consists of labeled 96x96 pixel image patches derived from large histopathology images.

The primary goal of this notebook is to understand the structure and distribution of the data.
We also verify image dimensions and format consistency. All findings will be documented here.
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
from PIL import Image

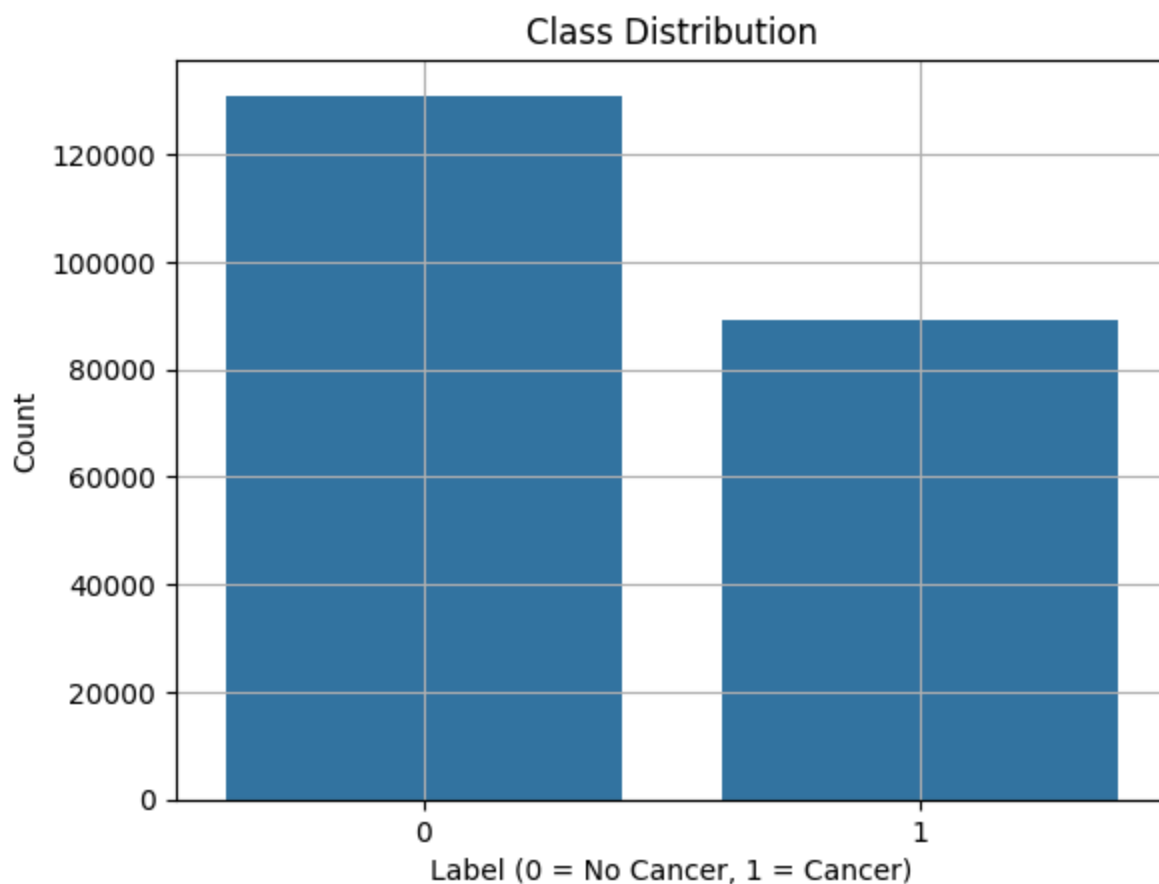
df = pd.read_csv("../data/train_labels.csv")
print("Shape:", df.shape)
df.head()
```

Shape: (220025, 2)

```
Out[16]:
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	id	label
0	f38a6374c348f90b587e046aac6079959adf3835	0
1	c18f2d887b7ae4f6742ee445113fa1aef383ed77	1
2	755db6279dae599ebb4d39a9123cce439965282d	0
3	bc3f0c64fb968ff4a8bd33af6971ecae77c75e08	0
4	068aba587a4950175d04c680d38943fd488d6a9d	0

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In [17]: sns.countplot(data=df, x="label")
plt.title("Class Distribution")
plt.xlabel("Label (0 = No Cancer, 1 = Cancer)")
plt.ylabel("Count")
plt.grid(True)
plt.show()
```



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In [20]: sample = Image.open("../data/train/" + df.iloc[0]['id'] + ".tif")
print(sample.size, sample.mode) # Should be (96, 96) and "RGB"
```

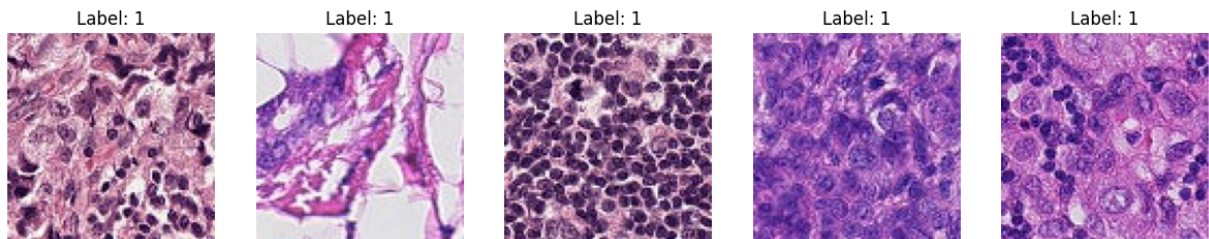
(96, 96) RGB

```
In [22]: from PIL import Image
import matplotlib.pyplot as plt

def show_samples(df, label, n=5):
    samples = df[df['label'] == label].sample(n)['id']
    fig, axs = plt.subplots(1, n, figsize=(15, 3))
    for i, img_id in enumerate(samples):
        img = Image.open(f"../data/train/{img_id}.tif")
        axs[i].imshow(img)
        axs[i].axis('off')
        axs[i].set_title(f"Label: {label}")
    plt.show()

show_samples(df, label=0)
show_samples(df, label=1)
```



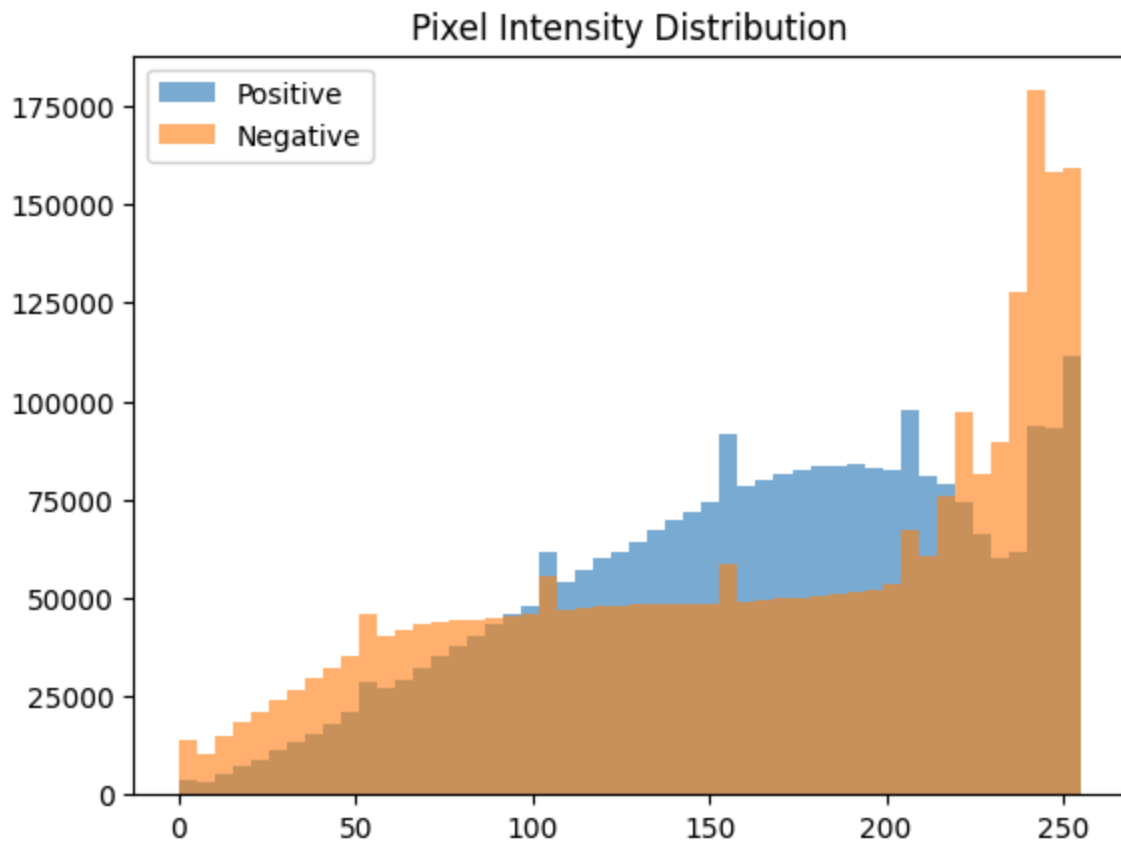


```
In [24]: import numpy as np

def pixel_histogram(img_ids):
    pixels = []
    for img_id in img_ids:
        img = np.array(Image.open(f"../data/train/{img_id}.tif"))
        pixels.append(img.flatten())
    return np.concatenate(pixels)

pos_ids = train_labels[df['label'] == 1]['id'].sample(100)
neg_ids = train_labels[df['label'] == 0]['id'].sample(100)

plt.hist(pixel_histogram(pos_ids), bins=50, alpha=0.6, label='Positive')
plt.hist(pixel_histogram(neg_ids), bins=50, alpha=0.6, label='Negative')
plt.legend()
plt.title("Pixel Intensity Distribution")
plt.show()
```



In []: